International Institute of Information Technology, Hyderabad

(Deemed to be University)

S24SC1.308 : The Universe Across Scales

Question cum Answer Booklet

Quiz 1, S-3

Total Marks: 16 Duration : 1 hour

16

Roll No. : 2022111004 Program : CSD Date : 30 JAN 2025

Room No. : SH 2 Seat No. : _____ Invigilator Signature : _____

Marks Table (to be filled in by the examiner)

Question No.	Marks								Examiner
1	5.5								
2	4.5								
3	4								
4	2								
5									

Instructions to the students:

1. This booklet consists of two sections: section 1 has five short answer-type questions and section 2 has four long answer-type questions. You are to answer all of the questions in section 1. Section has some optional questions, please read carefully before answering.
2. Place your ID card on the desk for verification by the invigilator.
3. Reading materials, such as books, and electronic gadgets, like tablets, computers, mobile phones or calculators, are not allowed inside the classroom.
4. Borrowing writing materials or calculators from other students during the quiz is prohibited. If any student is found indulging in malpractice or copying during the exam, the student will be given an 'F' grade for the course.

1. a. Prove that the critical perlocation probability in the 1 d lattice is 1 [0.5]
   b. Plot $\Pi(p, L)$ for $L = 1$ and $L = 2$ [1]
   c. Give the mathematical expression for n_s (probability of an arbitrary site (occupied or not) being, say the left hand side (LHSthe) of an s-cluster in 1-d). Explain the terms. From their show that cut-off cluster size is scaled with $(p_c - p)^{-1}$, when p tends to p_c [1 + 1.5]
   d. Prove that $\sum_{s=1}^{\infty} n_s S = p$ [1.5]

a) $\pi(p,L)$ is the probability that all points in Lattice ~~of size L~~ infinite are occupied when the occupation prob is p.

$$\lim_{L\to\infty} \pi(p,L) = \lim_{L\to\infty} p^L = \begin{cases} 0 & p<1 \\ 1 & p=1 \end{cases}$$

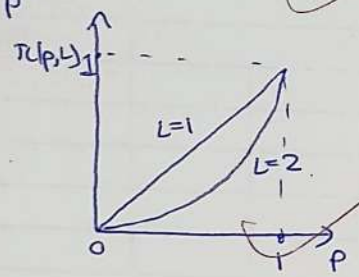
critical percolation prob p_c st $\lim_{L\to\infty} \pi(p,L) = \begin{cases} 0 & p<p_c \\ 1 & p\geq p_c \end{cases}$

comparing above two:

$p_c = 1$ — Ans.

0.5

b) $\pi(p,L) = p^L$



1

c) $n_s(p) \leftarrow$ prob that arbitary site (occupied or not) is the LHS of an s cluster.

$= (1-p)\, p^s\, (1-p)$ → RHS is not occupied.

↑ LHS is not occupied

p^s: this points occupied which form the cluster

} ensures cluster size is exactly s.

Direct prob multiplication is possible as we assume each ~~site's~~ site's occupance is independant.

$\therefore n_s = (1-p)^2 p^s$ — Ans.

$n_s(p) = (1-p)^2 p^s = (1-p)^2 \exp(\ln(p^s)) = (1-p)^2 \exp(s \ln(p))$ $\quad p_c = 1$ as shown in (a)

Let $S_\xi = \frac{-1}{\ln(p)}$ = cutoff cluster size.

$= \frac{-1}{\ln(p_c - (p_c - p))}$ as: $p_c - p \to 0$

$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} \dots$

as $x \to 0$

$\ln(1-x) = -x$.

using this.

$\Rightarrow \ln(p_c - (p_c - p)) = -(p_c - p)$

$\Rightarrow S_\xi = \frac{-1}{-(p_c - p)} = (p_c - p)^{-1}$ (as $p \to p_c$) — Ans.

1+1.5

d) To show $\sum_{s=1}^{\infty} s n_s = p$.

we know, $n_s = (1-p)^2 p^s$

$$\Rightarrow \sum_{s=1}^{\infty} s n_s = \sum_{s=1}^{\infty} s(1-p)^2 p^s = (1-p^2) \sum_{s=1}^{\infty} s p^s$$

$$= (1-p^2) \cdot p \sum_{s=1}^{\infty} s p^{s-1}$$

$$= (1-p^2) \cdot p \sum_{s=1}^{\infty} \frac{d}{dp}(p^s)$$

$$= (1-p)^2 p \frac{d}{dp}\left(\sum_{s=1}^{\infty} p^s\right)$$

now, $\sum_{s=1}^{\infty} p^s = \infty$ GP sum with $a = p$ and $r = p$

$$= \frac{a}{1-r} = \frac{p}{1-p}$$

$$\frac{d}{dx}\left(\frac{a}{b}\right) = \frac{a'b - b'a}{b^2}$$

$$\Rightarrow \frac{d}{dp}\left(\sum_{s=1}^{\infty} p^s\right) = \frac{d}{dp}\left(\frac{p}{1-p}\right) = \frac{(1-p)\cdot 1 - p(-1)}{(1-p)^2}$$

$$= \frac{1-p+p}{(1-p)^2} = \frac{1}{(1-p)^2}$$

Now, $\sum_{s=1}^{\infty} n_s \cdot s = (1-p)^2 p \frac{d}{dp}\left(\sum p^s\right) = (1-p^2)^2 p \cdot \frac{1}{(1-p)^2} = p$. —— Hence proved.

1.5

2. If $P(k) = ck^{-\gamma}$, Find out the normalization constant. Derive the m_{th} moment from this density distribution. When will the second and third moments $< k^2 >$ and $< k^3 >$ diverge? [2 + 2.5 = 4.5]

$p(k) = ck^{-\gamma}$ as this is a probability dist, $\int_{k_{min}}^{\infty} p(k)dk = 1$ where k_{min} is the minimum k value.

$$\int_{k_{min}}^{\infty} p(k)dk = \int_{k_{min}}^{\infty} ck^{-\gamma} dk = 1$$

$$\Rightarrow c \int_{k_{min}}^{\infty} k^{-\gamma} dk = c.\left[\frac{k^{-\gamma+1}}{-\gamma+1}\right]_{k_{min}}^{\infty} = \frac{c}{-\gamma+1}\left(k^{-\infty} - k_{min}^{1-\gamma}\right)$$

$$= \frac{c}{-\gamma+1}\left(0 - k_{min}^{1-\gamma}\right) = 1$$

$k_{min} > 0$ diverges if $k_{min} \to 0$.

$$\Rightarrow c = \frac{-(\gamma-1)}{k_{min}^{1-\gamma}} = \frac{(\gamma-1)}{k_{min}^{1-\gamma}}$$

$$\boxed{c = (\gamma-1)\, k_{min}^{(\gamma-1)}}$$ —Ans.

2

$$m^{th} \text{ moment} = \int_{k_{min}}^{\infty} p(k) k^m dk$$

$$= \int_{k_{min}}^{\infty} ck^{-\gamma+m} dk = c\left.\frac{k^{-\gamma+m+1}}{-\gamma+m+1}\right]_{k_{min}}^{\infty}$$

Divergent for $\gamma < m+1$.

Exists only for $\gamma \geq m+1$. as value is +ve else divergent. when value is −ve

$$= \frac{c}{-\gamma+m+1}\left(-k_{min}^{-\gamma+m+1}\right)$$

$$= \frac{-(\gamma-1)\, k_{min}^{(\gamma-1)}\, k_{min}^{-\gamma+m+1}}{-\gamma+m+1} = \frac{(\gamma-1)\, k_{min}^{m}}{(-\gamma+m+1)}$$

2

$$\langle k^m \rangle = \frac{(\gamma-1)\, k_{min}^{m}}{(-\gamma+m+1)} \quad \text{for } \gamma > m+1$$ —Ans.

Divergent when $\gamma \leq m+1$ as $\langle k^m \rangle$ becomes −ve or undefined.

$\therefore$ 2nd moment → $\gamma \leq 3$

3rd moment → $\gamma \leq 4$.

—Ans

2.5

3. Show that for a Bethe lattice with coordination number $z = 5$. Find out the ratio of the number of surface sites with respect to total sites. [4]

Each site connected to $z=5$ other sites
leads to $(z-1)=4$ sub branches

Let g - generation be the gth circle around the center (~~also~~ arbitrary)

$\therefore$ $g=1$ contains 5 sites $= z$

$g=2$ contains $z(z-1)$ sites $= 5.4$

$g=k$ contains $z(z-1)^{g-1}$ sites $= 5.4^{g-1}$

$g=0$ contains the one center point.

Total no. of sites till gth generation

$= \text{sum of } \{g=0, g=1, \ldots g=g\}$

$= 1 + 5 + 5.4 + \ldots 5.4^{g-1} = 1 + 5(1+4+4^2+\ldots 4^{g-1})$

$$a + ar + ar^2 + \ldots ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

here, $a=1$, $r=4$, $n=g$.

$$\therefore 1+4+4^2+\ldots 4^{g-1} = \frac{(1-4^g)}{1-4} = \frac{4^g-1}{3}$$

$\therefore$ Total sites till gth generation $= 1 + \frac{5}{3}(4^g-1)$ — ①

no. of surface sites at g^{th} generation $= 5.4^{g-1}$ — ②

$$\frac{②}{①} = \frac{\text{surface sites}}{\text{total sites}} = \frac{5.4^{g-1}}{1+\frac{5}{3}(4^g-1)} = \frac{15.4^{g-1}}{3+5.4^g-5} = \frac{15.4^{g-1}}{5.4^g-2} = \frac{15}{5.4 - \frac{2}{4^{g-1}}}$$

as $g \to \infty$

$$\lim_{g\to\infty} \frac{15}{5.4 - \frac{2}{4^{g-1}}} = \frac{15}{5.4} = \frac{15}{20} = \frac{3}{4}$$ — Ans.

4

4. Show that for the Bethe lattice, mean cluster size diverges at $p = p_c$. $(z=3)$ [2]

Let S = mean cluster size of a site

T = mean cluster size of one branch.

if site is not occupied (with prob $(1-p)$) contribute 0

else (with prob p) contribute $1 + (z-1)T = 1+2T$

$\therefore \; T = (1-p)\cdot 0 + p(1+2T)$

$T = p + 2pT$ also $S = 1 + 3T$

that point each branch.

$\Rightarrow T = \dfrac{p}{1-2p}$

$$S = 1 + 3T = 1 + \frac{3p}{1-2p} = \frac{1-2p+3p}{1-2p} = \frac{1+p}{1-2p} = \frac{1+p}{2(\frac{1}{2}-p)}$$

To calculate p_c.

each site can branch to $z-1$ subbranches

if ∞ cluster is found, then $p_c(z-1) = 1$

$\Rightarrow p_c = \dfrac{1}{z-1} = \dfrac{1}{3-1} = \dfrac{1}{2}$

$$S = \frac{1+p}{2(\frac{1}{2}-p)} = \frac{1+p}{2(p_c-p)} \quad \text{as } p \to p_c, S \to \frac{1+p}{0}$$

$\therefore$ as $p \to p_c$, S (mean cluster size) diverges.

— Hence proved.

2